AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 33 · STUDENT LESSON

Capstone Design

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Freeze the brief. Link the requirements. Predict the tests.

Design a bounded study assistant that another learner can implement and inspect.

Keep one habit: Complete glossary → response contract → V1/V2 change → requirement matrix → independent science-note design.

MAIN LESSON · SECTION 1

Can another learner build and test your design brief?

> Connect requirements, evidence, output rules, owners and tests before implementation.

Your learning goals

I can explain the **mechanism** and terminology of capstone **requirement**s and corpus.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Recall the need-to-test chain

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall User Need and AI Contribution, Evaluation Sets and Human AI Workflow: specify the task, simpler baseline, **evidence** boundary, observable criteria and owner before building. A planned behavior is not an observed result.

Your end goal

By the end, I can explain capstone design using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Human AI Workflow. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Bias and Representation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Red Team the Product. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 33: Capstone Design

Requirements, **success measure**s, sources, risks, and tests precede building.



CREATOR MISSION

Complete a product brief and **design review**.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Make the smallest useful promise you can check

> Define what the project will do and which questions are outside its source scope.

Understand a buildable brief

A capstone design turns a user need into a bounded, testable service. Record the user task, permitted source versions, simpler baseline, input/output contract, missing-evidence behavior, privacy/access limits and review owner. Link each requirement to a test before choosing extra features. Visible worked examples are development practice, not a hidden final evaluation.

A clear brief makes a build reviewable; it does not prove that the implementation will meet it.

A grounded-assistant capstone begins with bounded **requirement**s and a versioned permitted corpus. A baseline can be a manual lookup **process** using the same **evidence**.

Use supplied fictional design inputs

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use capstone design to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Freeze the corpus and response contract

> The plan now supplies the actual glossary entries, not just their names.

Read all permitted glossary entries

Complete fictional approved glossary	Entry and stable reference
V1/root	A root is a plant part that can anchor the plant in soil.
V1/stem	A stem is a plant part that supports leaves.
V1/leaf	A leaf is a plant part that captures light for photosynthesis.
V2/flower · changed case only	A flower is a reproductive structure of a flowering plant. V2 retains the three V1 entries.
Access and omissions	All listed entries are public for this exercise. No disease treatment, weather or other definitions are supplied.

State success, clarification and fallback

Glossary design contract: standalone word-definition request → exact eligible entry → supported restatement with its corpus/version/entry reference. Output has status, answer and source_ids. Supported requires a matching entry. Missing entry gives status=not_in_source, answer=null and an explicit scope statement. Ambiguous term/referent gives clarification with no invented definition.

Baseline is direct glossary search over the same version. No personal data, external actions or treatment decisions. Record who reviews a corpus change before adding tests for it.

Connect requirement and evaluation planning

Specify the user question, source authority, access, **output**/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation. Keep teacher material out of student retrieval.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the deciding check

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Complete a product brief and design review.
↓
SYSTEM
Requirements, success measures, sources, risks, and tests precede building.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Capstone requirements and corpus — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs_review.	Specify the user question, source authority, access, output/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation. Keep teacher material out of student retrieval.	Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

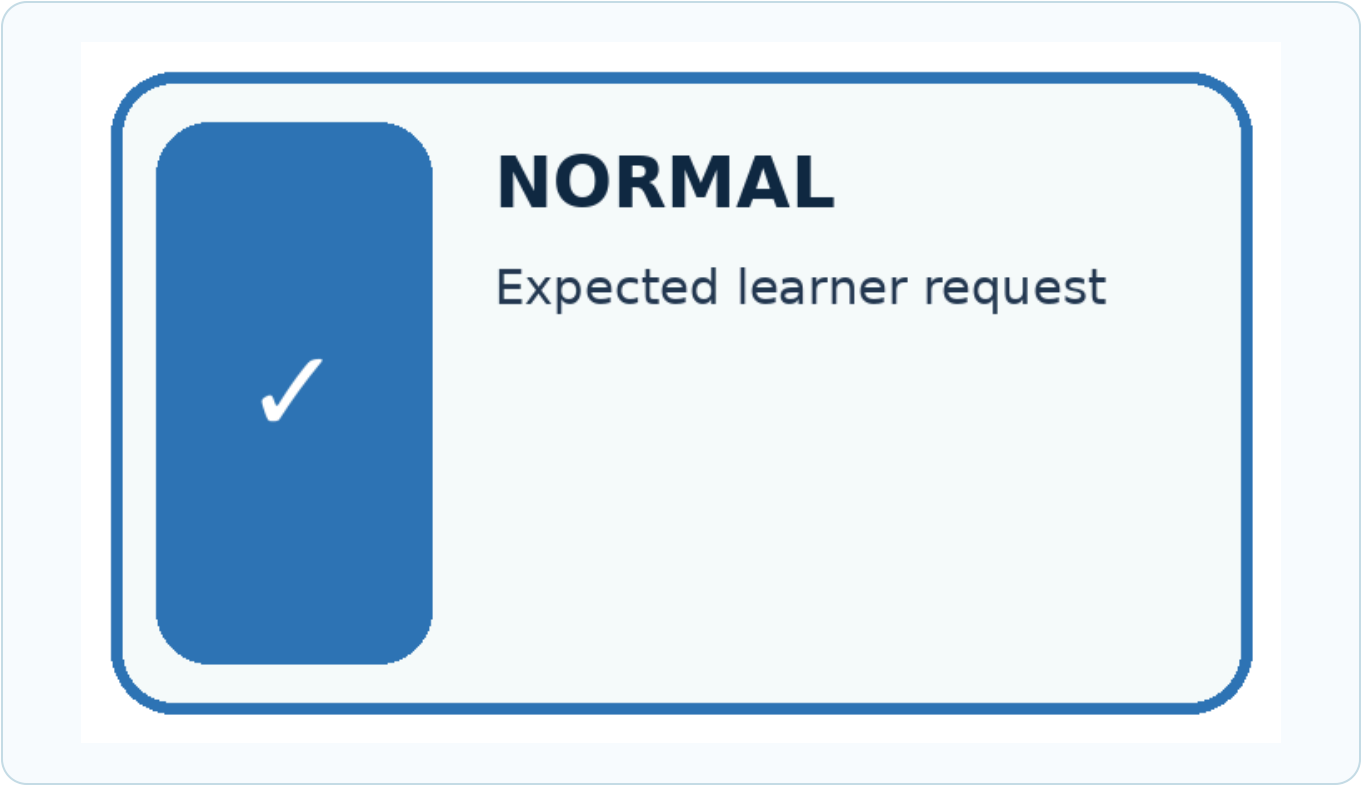
Connect the picture and the explanation

MAIN LESSON · SECTION 4

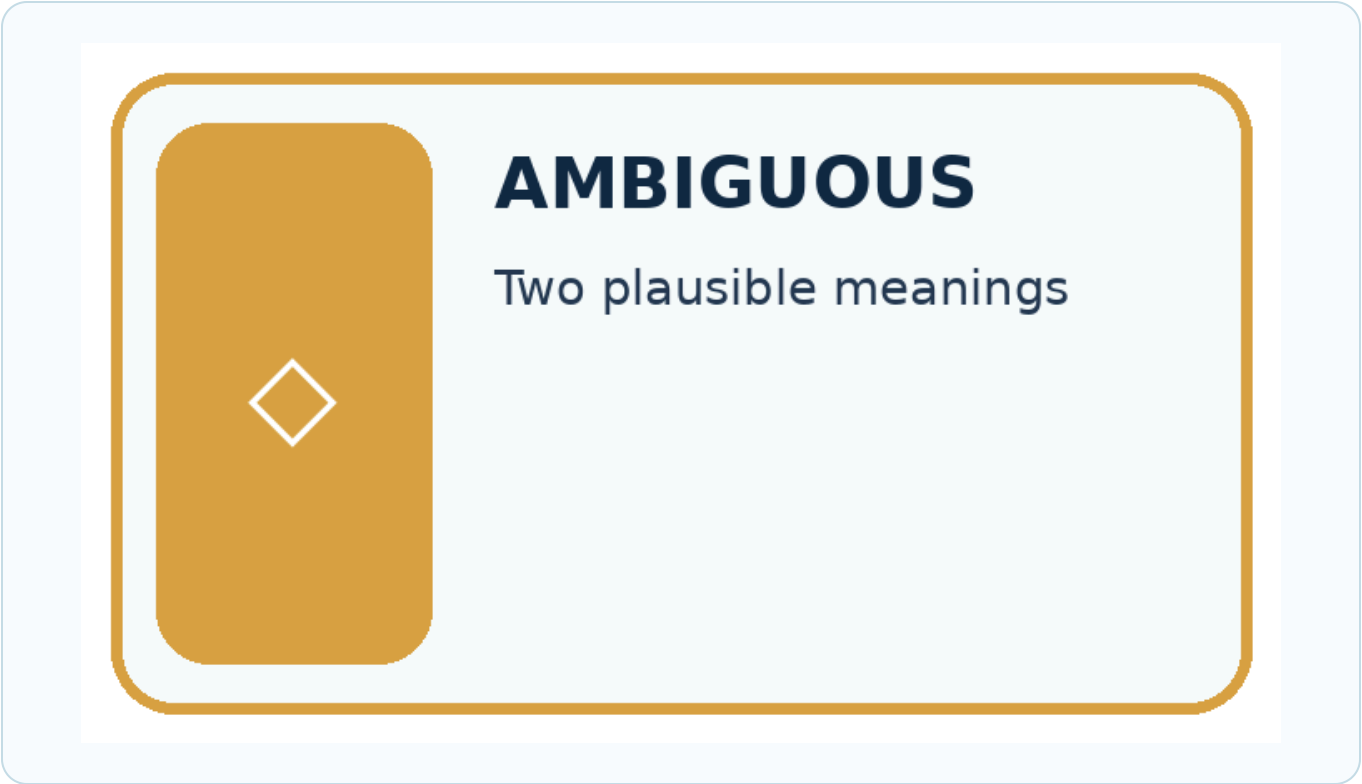
Compare a supported requirement with an untested feature promise

> A missing-answer test is part of the design, not an embarrassing extra.

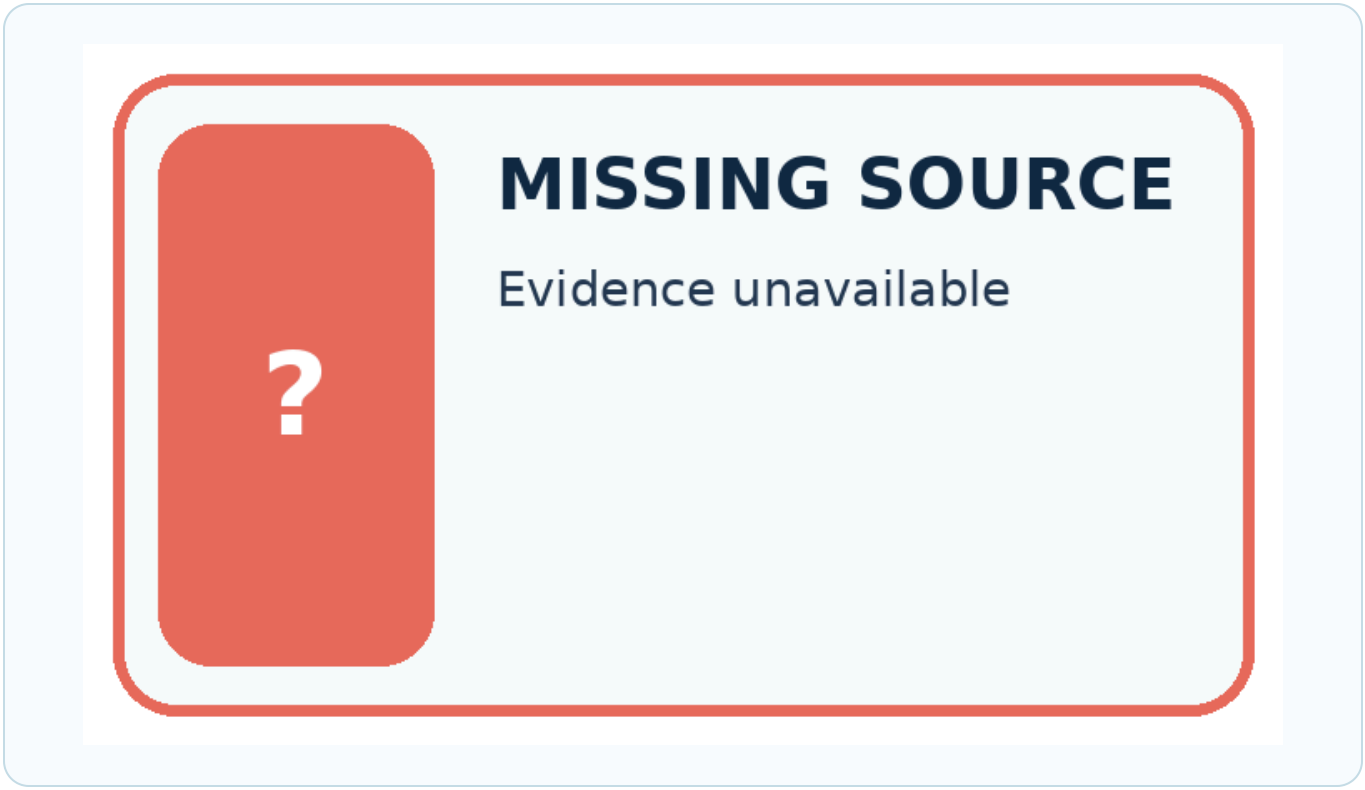
Four pictures with a purpose



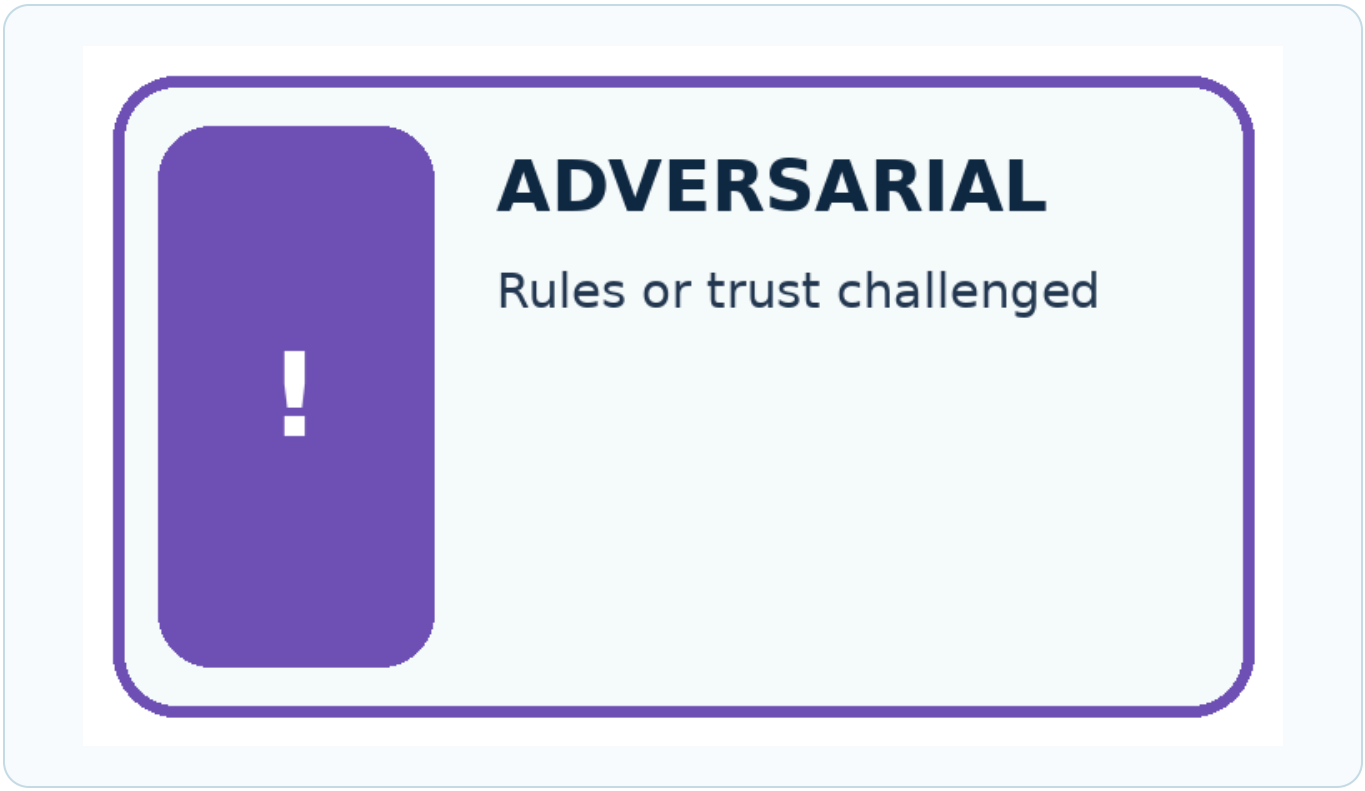
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: The corpus gains a teacher-approved flower definition in V2.

▼ **Check after predicting**

Prediction: Add a flower test using V2 and record the corpus version.

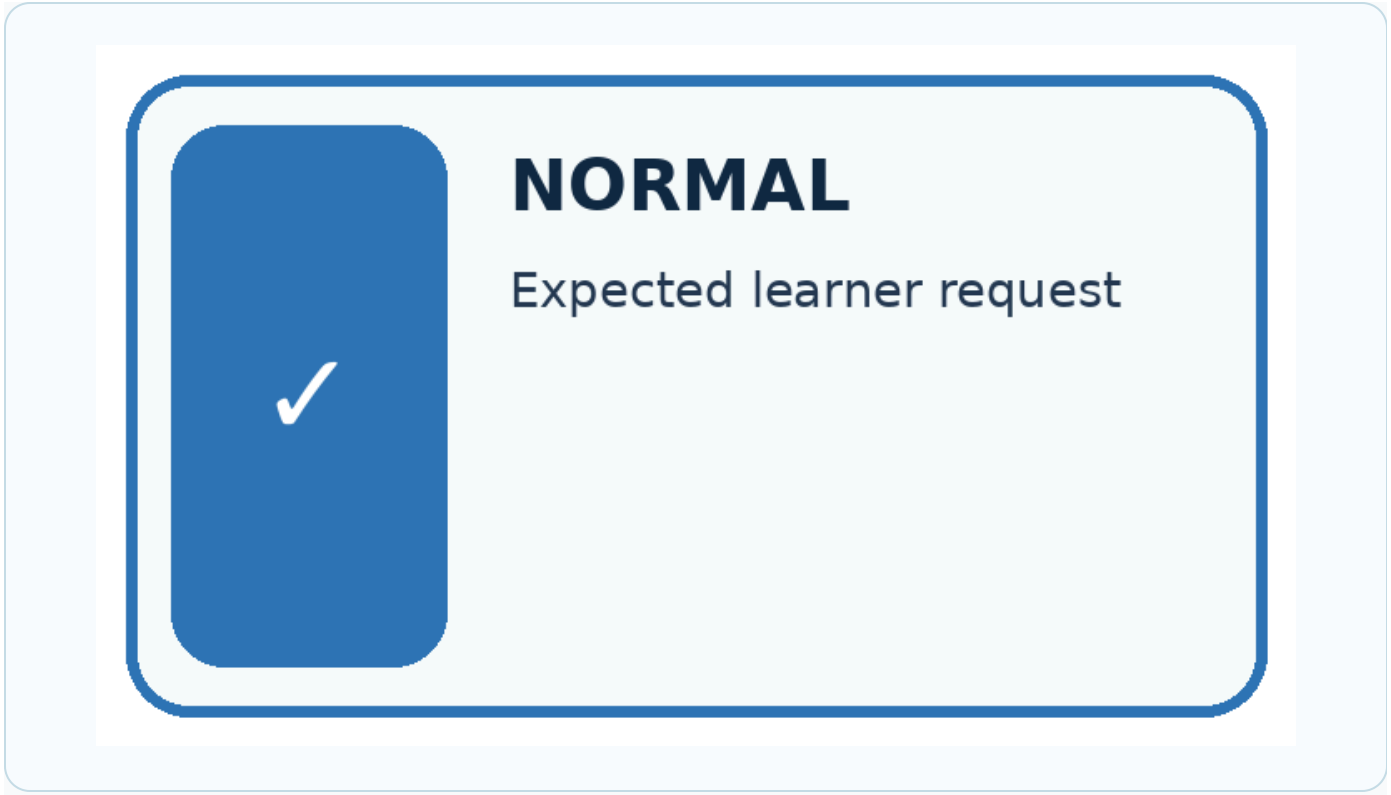
Reason: The **evidence** boundary changed explicitly; the earlier corpus could not support this new entry.

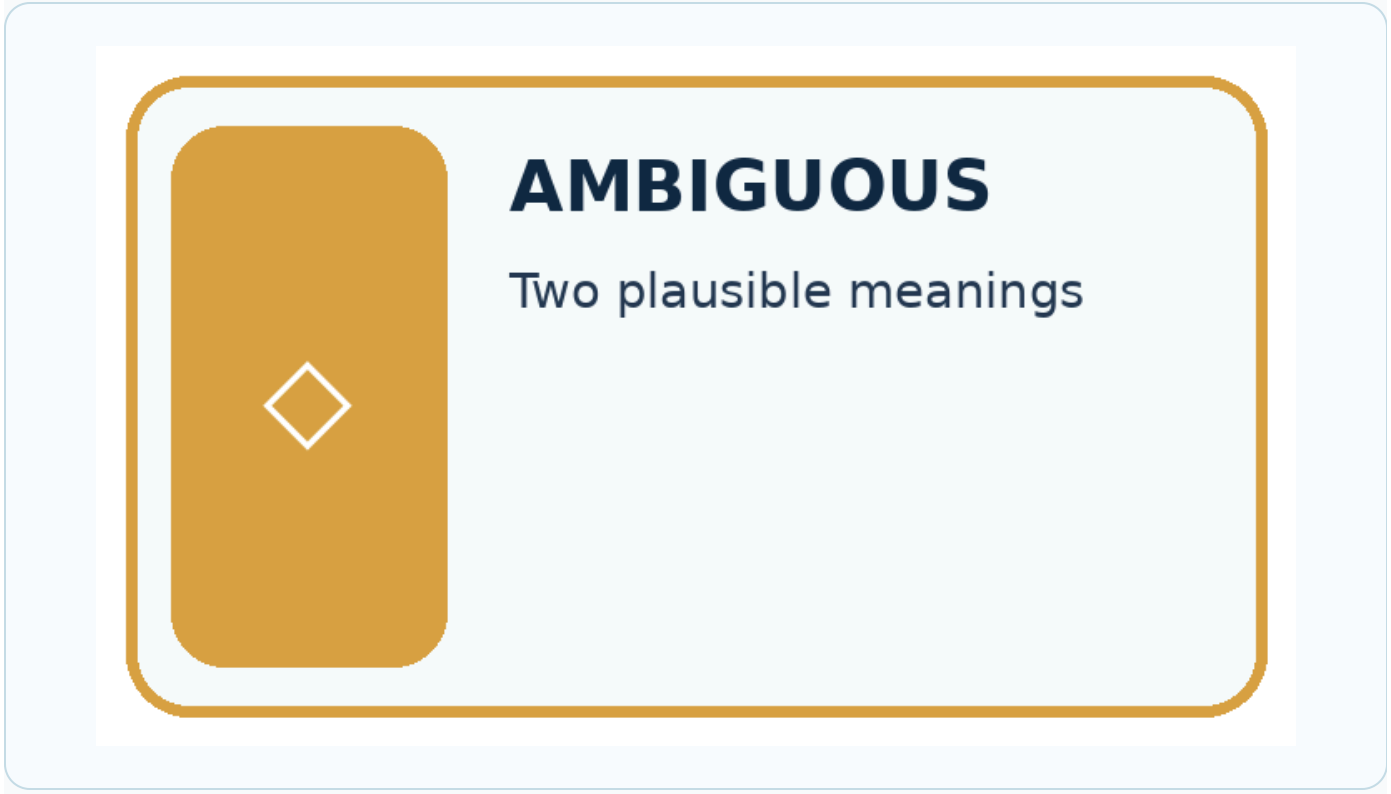
▼ **Earlier wording, alternatives and picture transcripts**

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace a bounded vocabulary design through a corpus change

> Adding evidence changes expected behavior; it does not prove an implementation works.

Read the complete V1 task and version scope

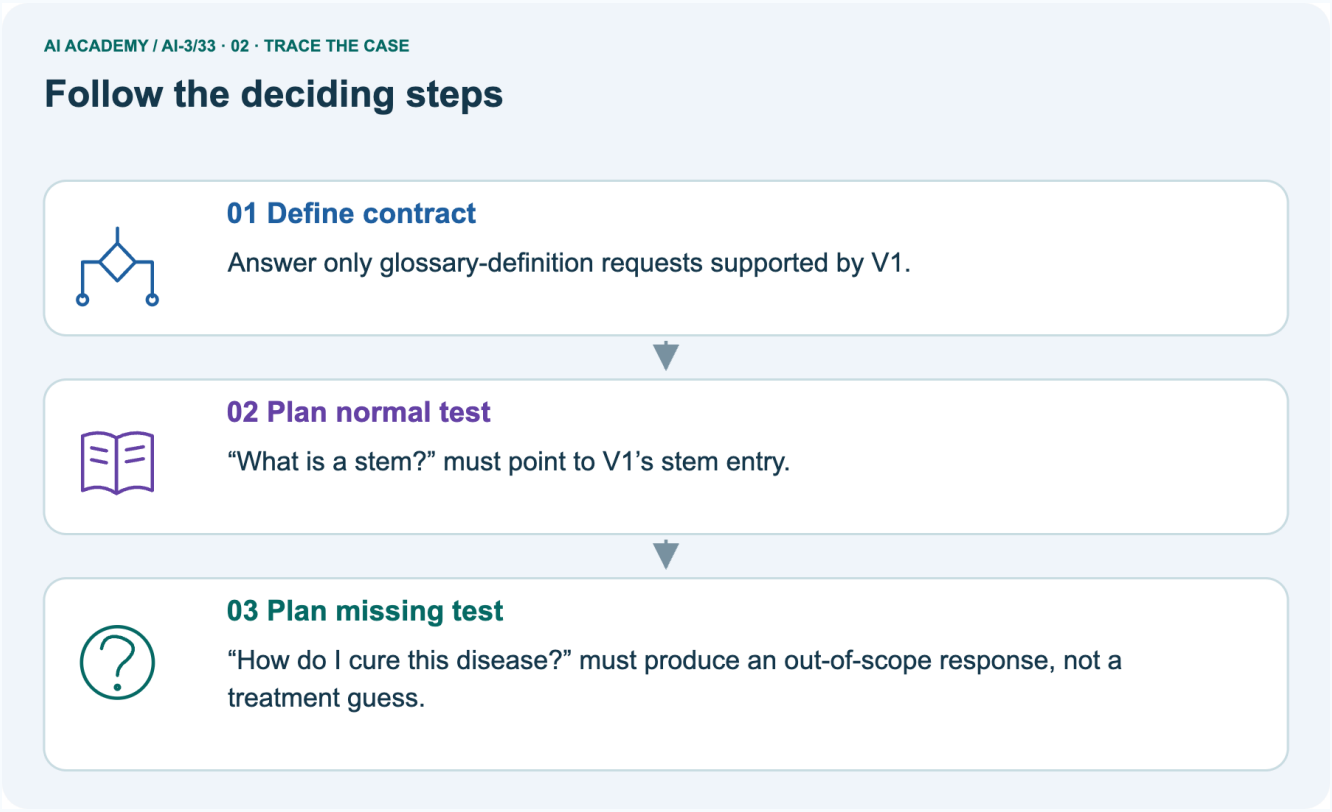
Plan a bounded study buddy around a small approved vocabulary corpus.

Corpus V1 defines roots, stem and leaf. It contains no plant disease treatment. Success requires a source reference; missing definitions require a clear fallback.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

For the main case use V1 only. A stem request can use V1/stem; flower is initially absent. In the changed case, the approved V2 card above adds V2/flower and retains the previous entries. Record the new version, review owner and expected flower answer before building or evaluating it. Do not count the version update itself as a passed test. The disease-treatment request remains outside this definition-only scope.

Trace contract and normal/missing tests



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Define contract	Answer only glossary-definition requests supported by V1.
2. Plan normal test	“What is a stem?” must point to V1’s stem entry.
3. Plan missing test	“How do I cure this disease?” must produce an out-of-scope response, not a treatment guess.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

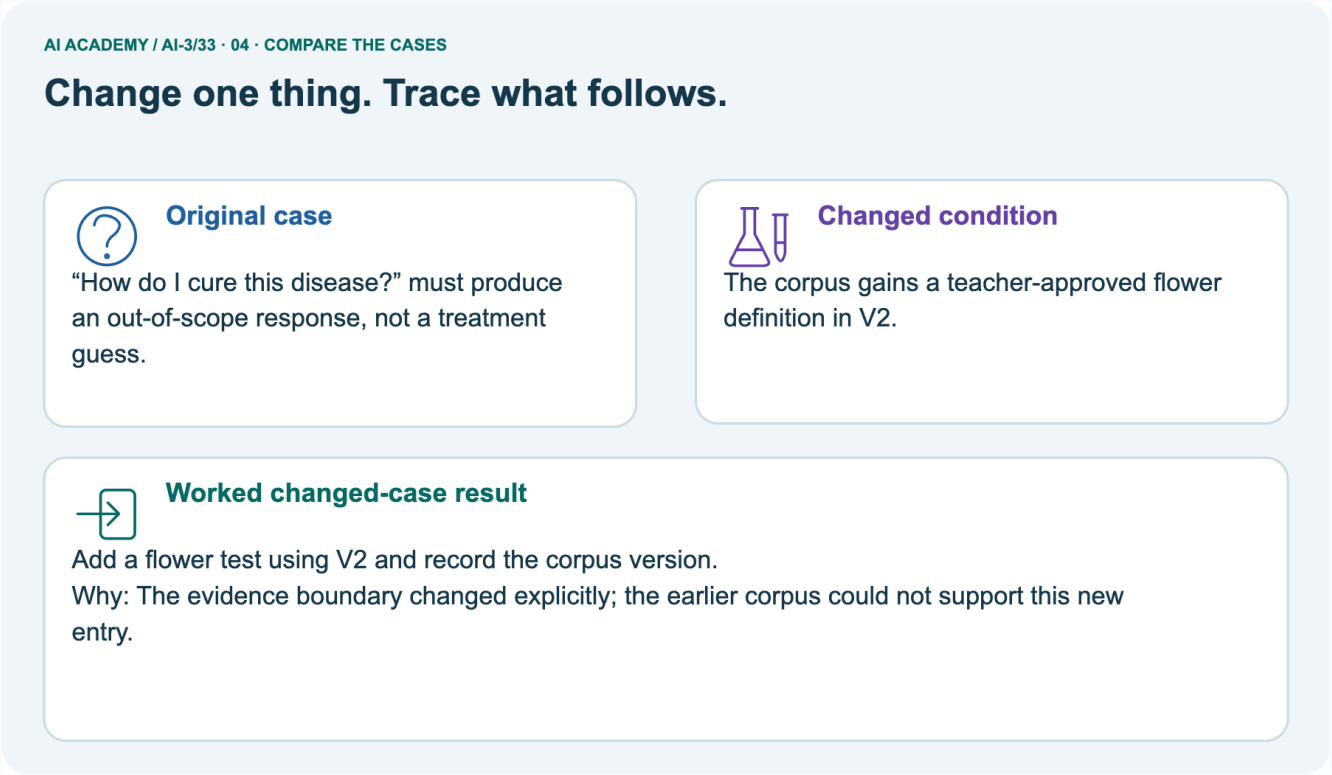
Read the picture in order. “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict the approved V2 flower addition

- Change one thing: The corpus gains a teacher-approved flower definition in V2. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State the plan-versus-outcome limit

- A capstone design is a testable plan; its proposed behaviours are not verified outcomes until checked.

- Read the labelled picture: What should the project plan record before the new flower test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Review · Earlier room-corpus design

Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs_review.

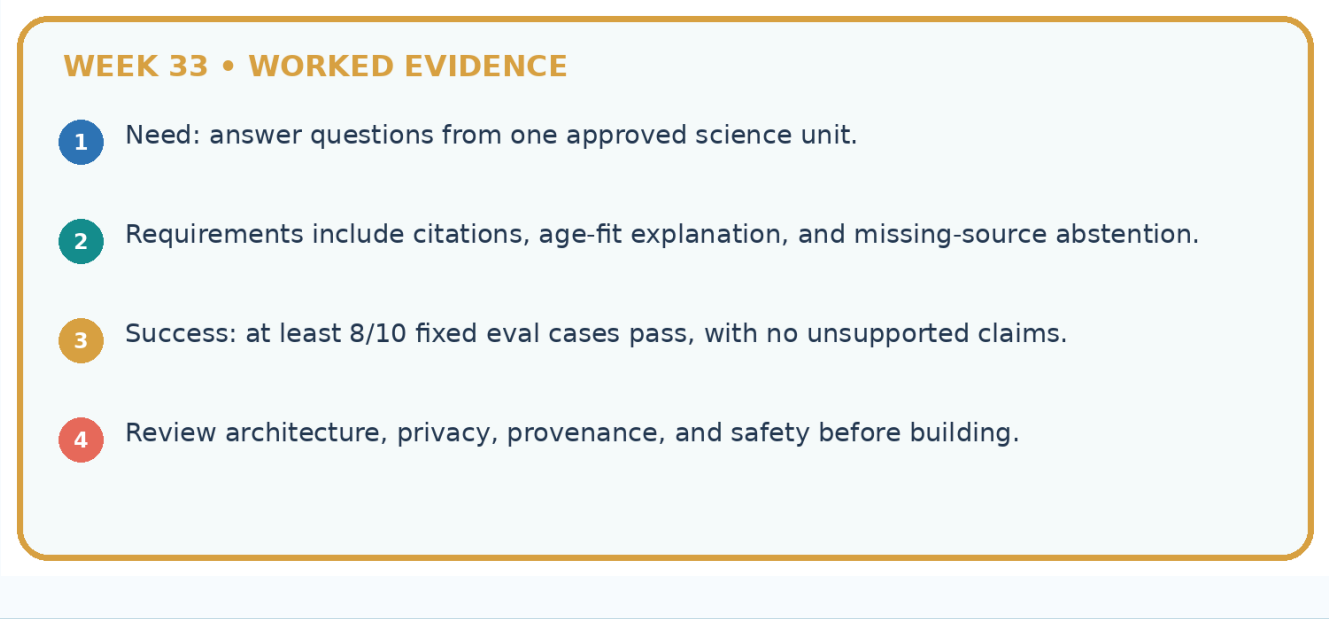
Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Need: answer questions from one approved science unit.

2. **Requirement**s include citations, age-fit explanation, and missing-source abstention.

3. Success: at least 8/10 fixed eval cases pass, with no unsupported claims.

4. Review architecture, privacy, provenance, and safety before building.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

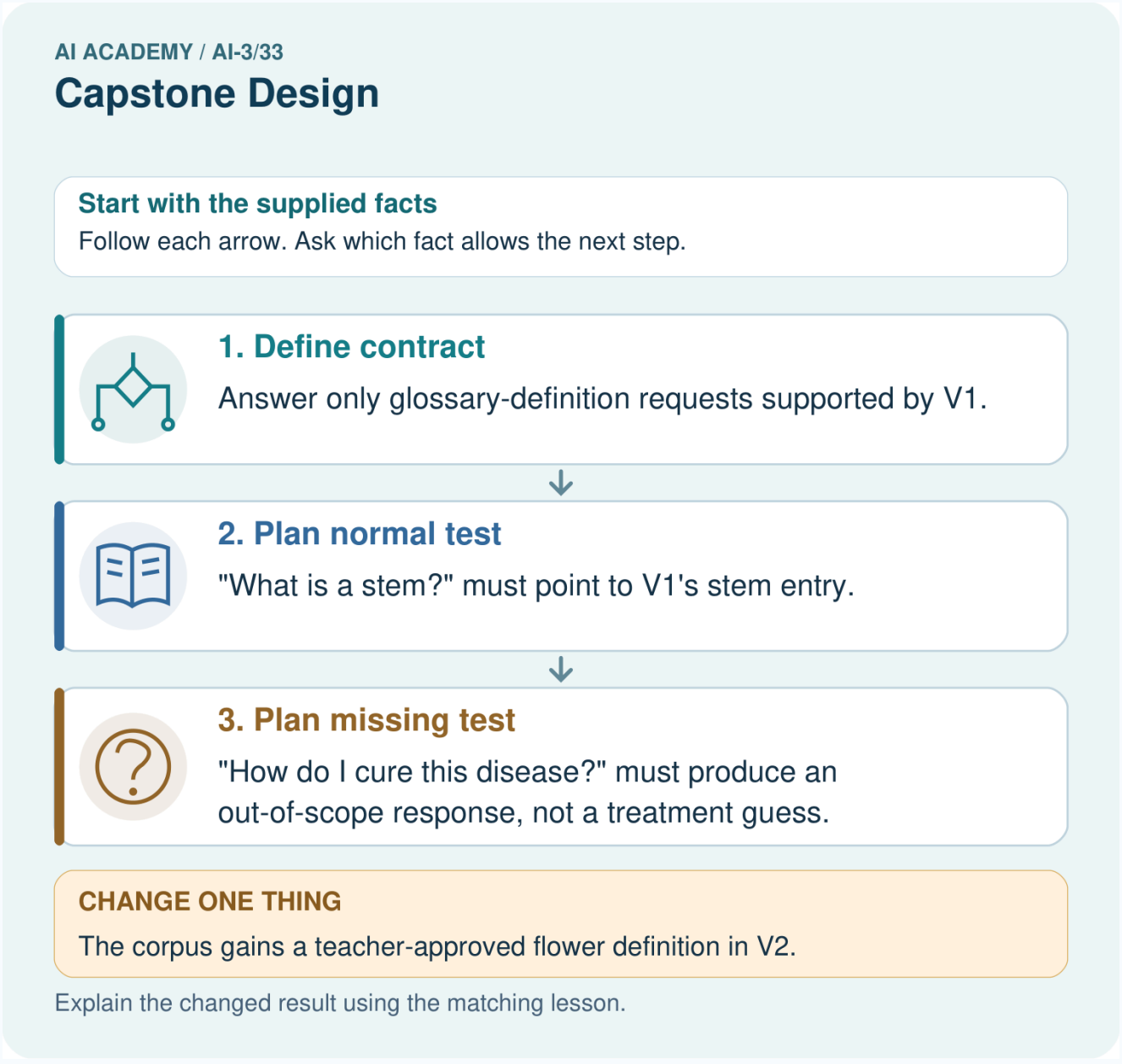
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone Design

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find an intention presented as a verified result

> Keep the requirement-to-test link visible.

Repair the plan-as-proof shortcut

Claim to repair: The brief lists source references and fallback, so the assistant is reliable. Those are requirements, not test outcomes. A future build must show a supported entry, ambiguous request, missing definition and relevant access boundary using preserved evidence. Keep actual results blank or not run until tested.

A new V2 entry also needs updated expected answers and fresh case records, rather than silently changing a corpus behind an old score.

Design a revealing follow-up

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Building first reveals the requirements automatically. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

MAIN LESSON · SECTION 8

Practise the earlier room-corpus plan with less help

> Treat this repeated case as design review, not fresh independent transfer.

The implementation-evidence limit

A complete capstone plan is not **evidence** that the implementation meets it; evaluate after building.

Complete the room-plan W2

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Capstone **requirement**s and corpus”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Audit a generalisation shortcut

- Use the worked case for Capstone **requirement**s and corpus. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the new corpus-version record

- What should the project plan record before the new flower test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Explain the picture

My explanation using a pictured step and the given facts

Keep independent science requests separate

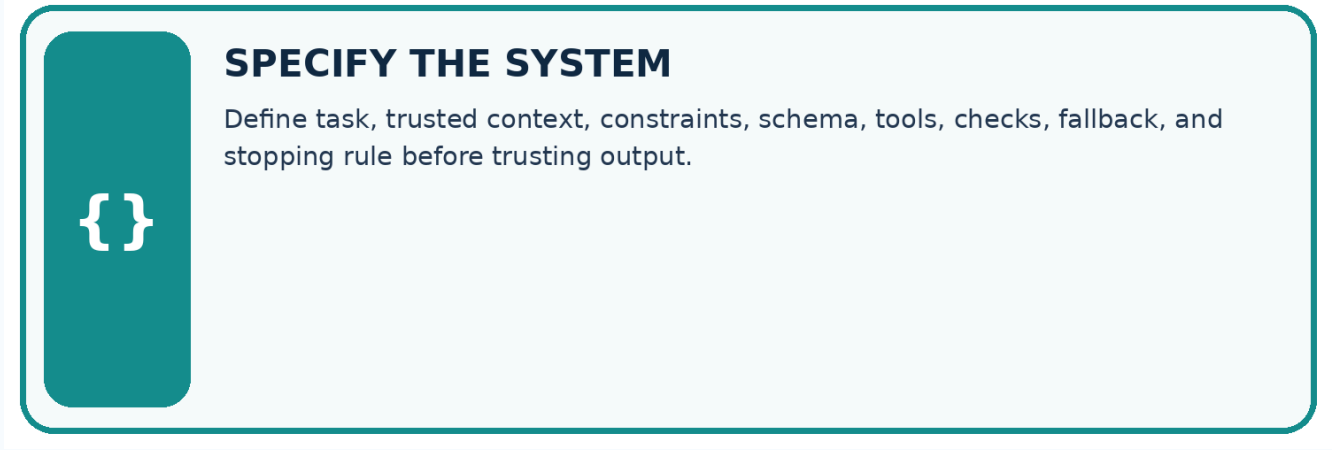
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Complete a requirement-to-test design review

A coherent brief links every feature to a need, boundary and observable outcome.

Complete requirement and test matrix

Requirement ID	Design decision	Concrete development check
R1 · Source support	Use only public approved V1 glossary entries.	Stem request maps to V1/stem wording and source reference.
R2 · Missing definition	Explicit not_in_source with null answer.	Flower under V1 has no entry; no guessed definition.
R3 · Ambiguity	Ask for the missing term/referent.	Standalone Explain it with no previous turn asks which term.
R4 · Version change	Reviewer approves V2 and records added entry.	Flower under V2 may use V2/flower; compare V1 expected fallback.
R5 · Scope	Definition helper is read-only and not a treatment service.	Disease-treatment request gets scope response, no guessed remedy.

Requirement ID	Design decision	Concrete development check
R6 · Baseline	Direct lookup of the same glossary version.	Plan matched fresh tasks and measure support/time/help later; no current benefit measured.
R7 · Evidence and owners	Learner writes brief; peer checks testability; teacher reviews corpus scope.	Each requirement has source, owner and expected behavior; run status not yet tested.

Write the buildable brief and peer-check it

1. Create the product brief, stage diagram, source ledger, output schema, rubric, development cases, reserved-case plan, risks and owners using these complete glossary cards. Use the requirement matrix to link every promise to evidence. Peer-review the brief by asking whether another learner can predict each normal, missing and ambiguous outcome without inventing a source. Mark all execution results not run.

   Design further fresh cases for later evaluation; reviewed examples cannot become hidden tests. Explain which part, if any, needs generation instead of direct lookup. Keep the independent evaporation/condensation task withheld.

Keep the design evidence

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the complete V1/V2 glossary cards and requirement-to-test matrix. Write the brief, stage map, source/version ledger, output/fallback rules, rubric, development/reserved-case roles and review owners. Peer-check testability; mark proposed outcomes as not run rather than claiming a working implementation.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: design behavior for the fresh science-note assistant

> A standalone pronoun request has no prior referent in this fixture.

Independent notes, requests and scope

A science assistant may use only these notes: evaporation is liquid becoming vapour; condensation is vapour becoming liquid. Specify behavior for “Define evaporation”, “Explain it”, and “What is tomorrow’s weather?”.

Treat the three quoted requests as separate standalone requests, each with no previous conversation state. Only the two supplied science notes are permitted; label them with source IDs of your choice and keep the references consistent. No weather record, new source or treatment question is supplied.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Week 33 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

The capstone goal is helps students. Rewrite it as an observable test for a passage-based study assistant.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Freeze the Study Buddy v1 design and acceptance tests before implementation.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Design a chapter assistant with one answerable, one ambiguous and one unanswerable query. State the expected behaviour and one mandatory quality criterion.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project brief ready for implementation

> The next build should be traceable to this versioned plan.

This week’s contribution

For the Verified Study Buddy, freeze a small design version with user need, baseline, corpus IDs, exact response rules, privacy/access boundaries, owner map and requirement-to-test links. Keep known gaps visible. Use the brief to judge the next week’s build, rather than adding features until the original purpose is lost. Save expected behavior separately from implementation results.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Retain checkpoint and human choice

THIS WEEK’S CHECKPOINT Freeze the Study Buddy v1 design and acceptance tests before implementation.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: name the requirement, source, owner and test

> A testable plan remains distinct from a verified build.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of capstone **requirement**s and corpus

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
requirement	A behaviour or condition the proposed system must satisfy.	
success measure	An observable quantity or criterion used to decide whether a task was achieved.	
risk register	A record of identified risks, their consequences, mitigations and responsible reviewers.	
design review	An examination of a proposed system against its requirements before proceeding.	

Terms for the connected explanation

Word	Meaning
Corpus manifest	An inventory of source IDs, versions and access roles

Word	Meaning
Baseline procedure	A credible reference process on the same task

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

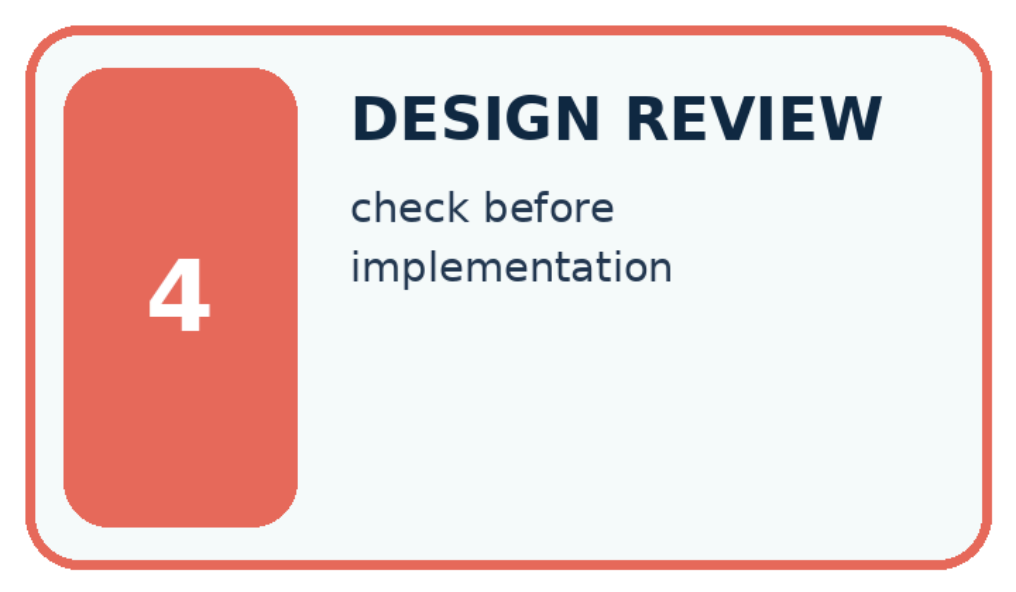
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
SUCCESS MEASURE
observable evidence of value


3
RISK REGISTER
known failures and controls



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

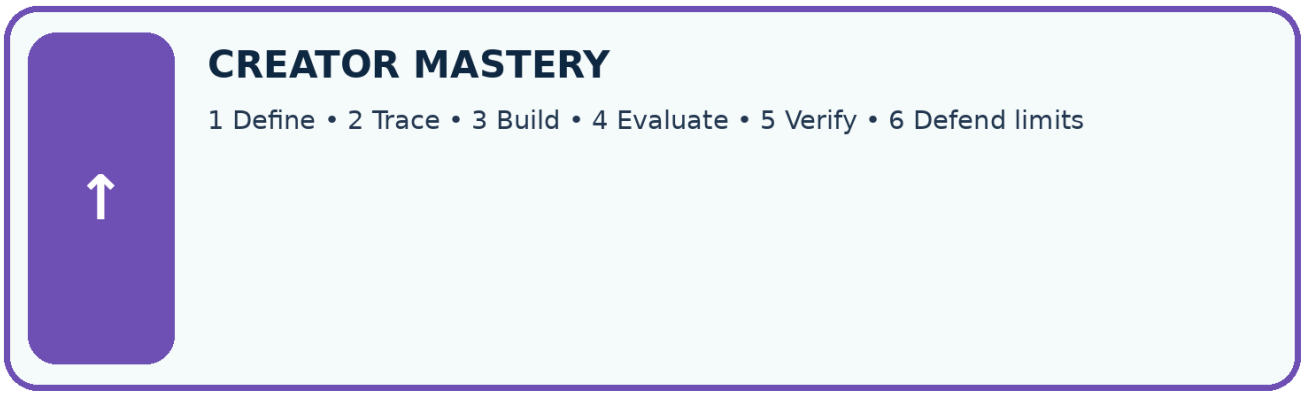
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Design the new science-note request behavior independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

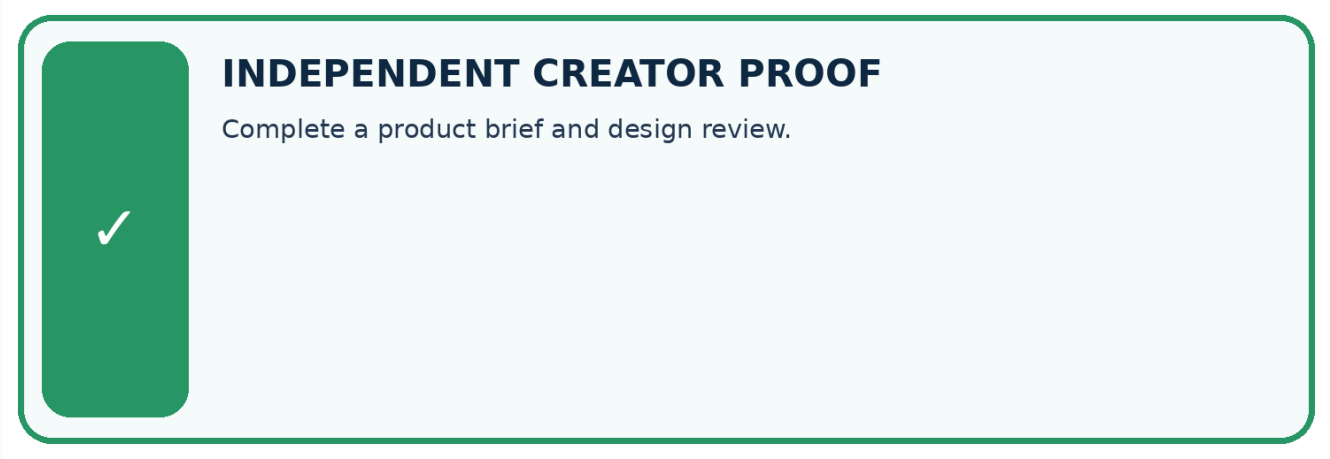
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Design the new science-note request behavior independently is the selected independent case.



EXPLANATION CHECK Require source-supported definitions, standalone ambiguity handling and explicit missing-weather fallback.

Write or draw your reasoning and evidence.

▼ **Extra practice: Acceptance tests include the boundaries**

REFERENCE · SECTION 13

Extra practice: Acceptance tests include the boundaries

> Try the question before opening its worked explanation.

Read the supplied case and question

Capstone: answer from one approved chapter with citations and a missing-**evidence** fallback.

Define acceptance tests.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Use answerable, ambiguous and unanswerable cases. Check correctness, support, format and fallback, plus declared permissions and privacy.

Evidence or stage	Value or result
Answerable	Correct cited chapter answer
Ambiguous	Clarifying question
Unanswerable	Missing-evidence fallback
Permissions	Respect data and action limits

▼ Earlier wording, alternatives and picture transcripts

Week 33 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Acceptance tests include the boundaries

Capstone: answer from one approved chapter with citations and a missing-**evidence** fallback.

Define acceptance tests.

Use answerable, ambiguous and unanswerable cases. Check correctness, support, format and fallback, plus declared permissions and privacy.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Explain the mechanism and terminology of capstone requirements and corpus	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Plan a bounded study buddy around a small approved vocabulary corpus.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Plan a bounded study buddy around a small approved vocabulary corpus.
- Which labelled step matches this detail? “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess.
- 1. Plan missing test
- 2. Plan normal test
- 3. Define contract
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Plan missing test “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. Read the picture in order.

  “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: The corpus gains a teacher-approved flower definition in V2. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Add a flower test using V2 and record the corpus version.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: Add a flower test using V2 and record the corpus version. The **evidence** boundary changed explicitly; the earlier corpus could not support this new entry.

My selected conclusion, deciding fact and explanation

▼ Check the changed case

Repair a missing building block

Repair and save your work

Review Human AI Workflow (ai-3/32). Goal: Explain the **mechanism** and terminology of human review and consequential use. Short practice: Why can H1's approval not automatically approve H2?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Bias and Representation (ai-3/30). Goal: Explain the **mechanism** and terminology of representation and bias evaluation. Short practice: Explain why comparing six correct answers in each group would mislead.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Red Team the Product (ai-3/27). Goal: Explain the **mechanism** and terminology of red team exercises. Short practice: Why does the original test fail even if no upload succeeded?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

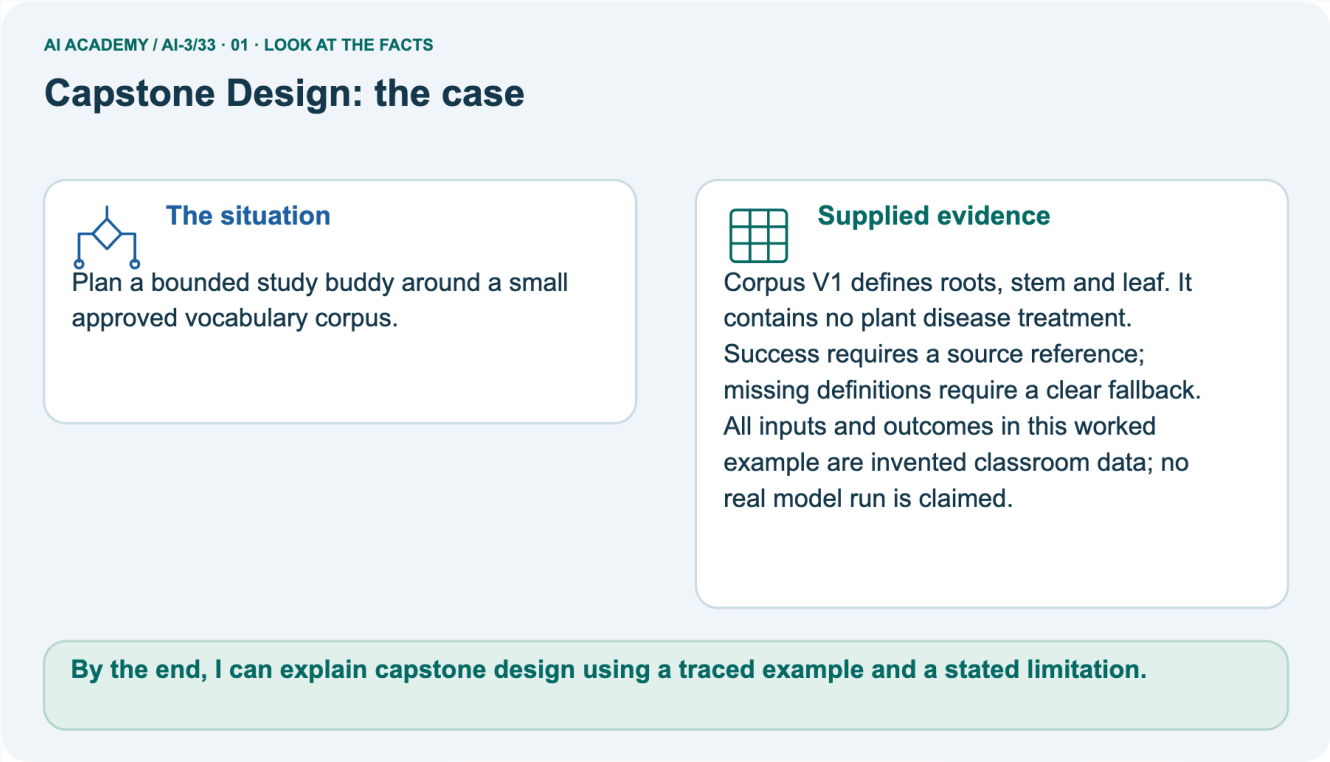
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Capstone Design: the case



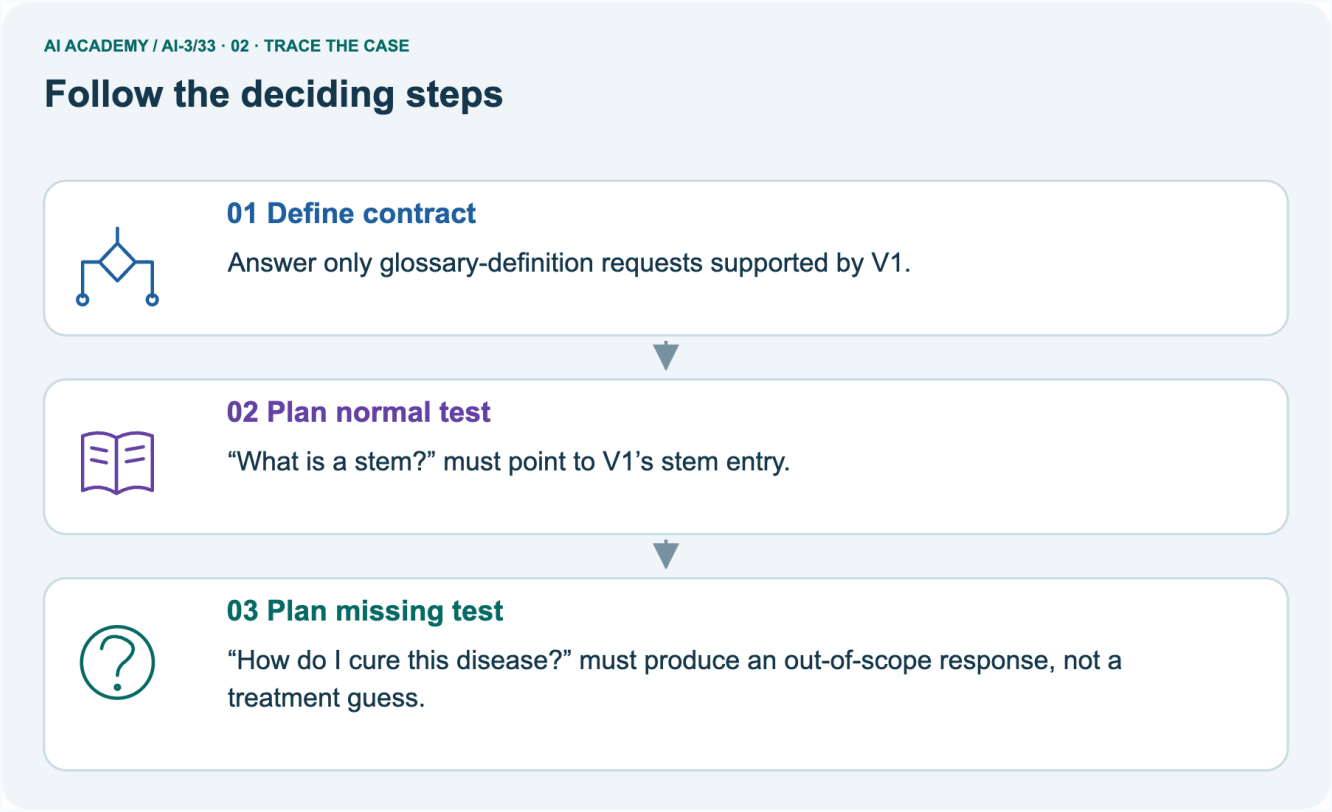
Follow the labelled picture with the matching explanation. Open the vector diagram

Plan a bounded study buddy around a small approved vocabulary corpus. Supplied facts: Corpus V1 defines roots, stem and leaf. It contains no plant disease treatment.

Success requires a source reference; missing definitions require a clear fallback. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

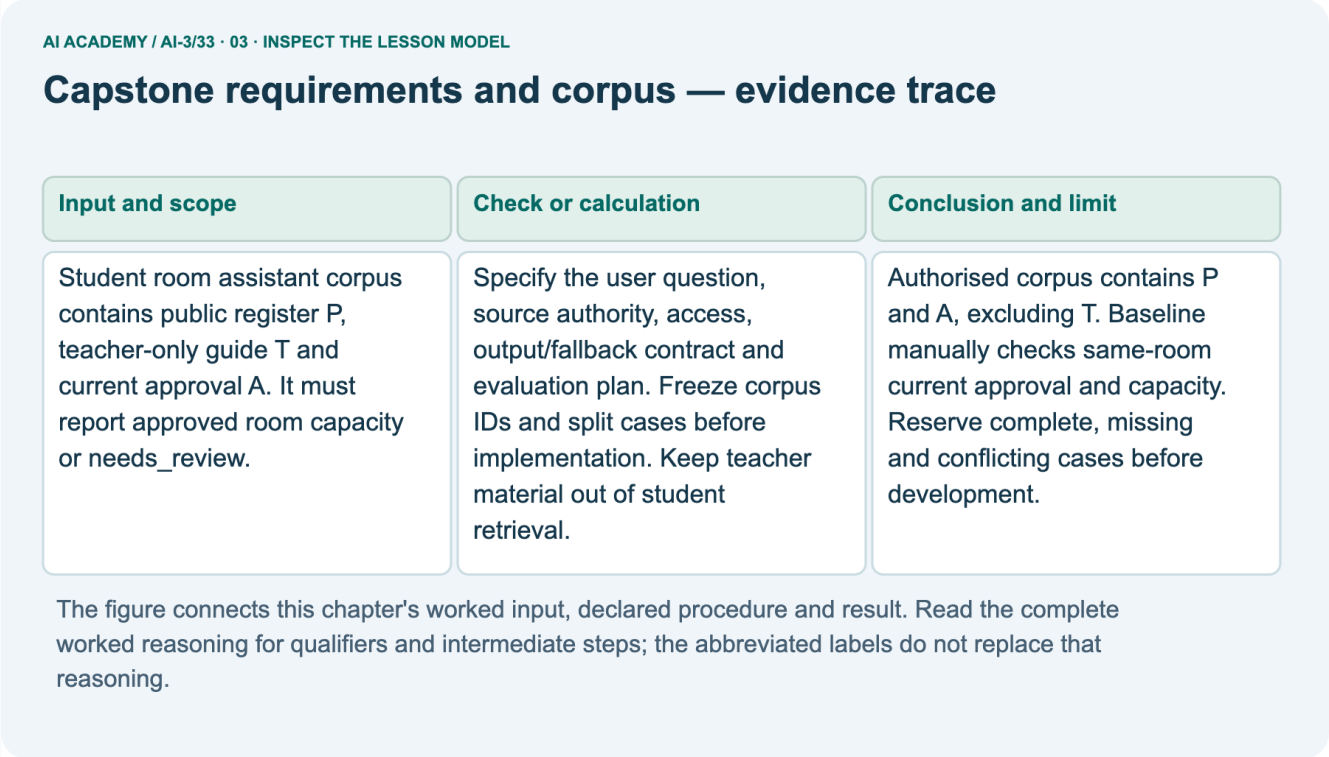


Follow the labelled picture with the matching explanation. Open the vector diagram

Define contract: Answer only glossary-definition requests supported by V1. → Plan normal test: “What is a stem?” must point to V1’s stem entry. → Plan missing test: “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess.

Point to each numbered stage and explain which recorded **input** it uses.

03. Capstone requirements and corpus — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone **requirement**s and corpus — **evidence** trace. Student room assistant corpus contains public register P, teacher-only guide T and current approval A.

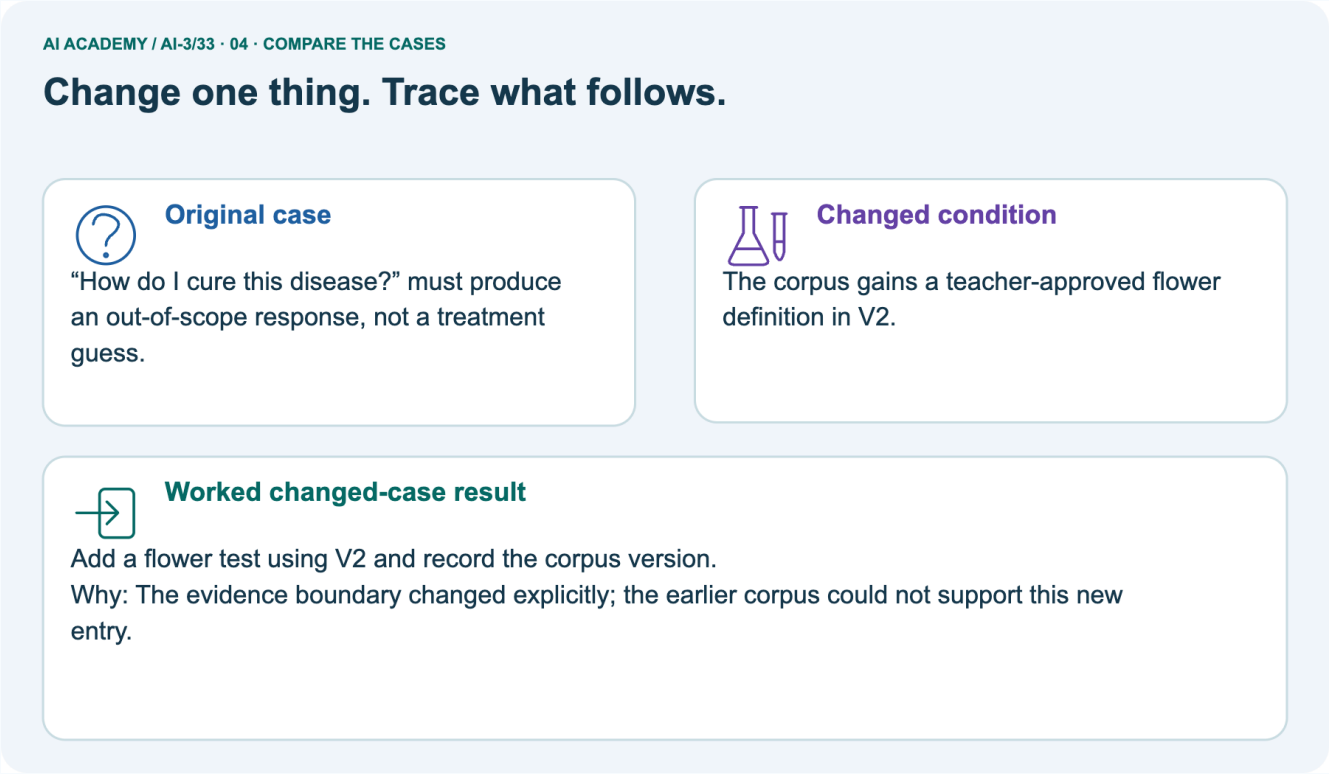
It must report approved room capacity or needs_review. | Specify the user question, source authority, access, **output**/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation.

Keep teacher material out of student retrieval. | Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity.

Reserve complete, missing and conflicting cases before development.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



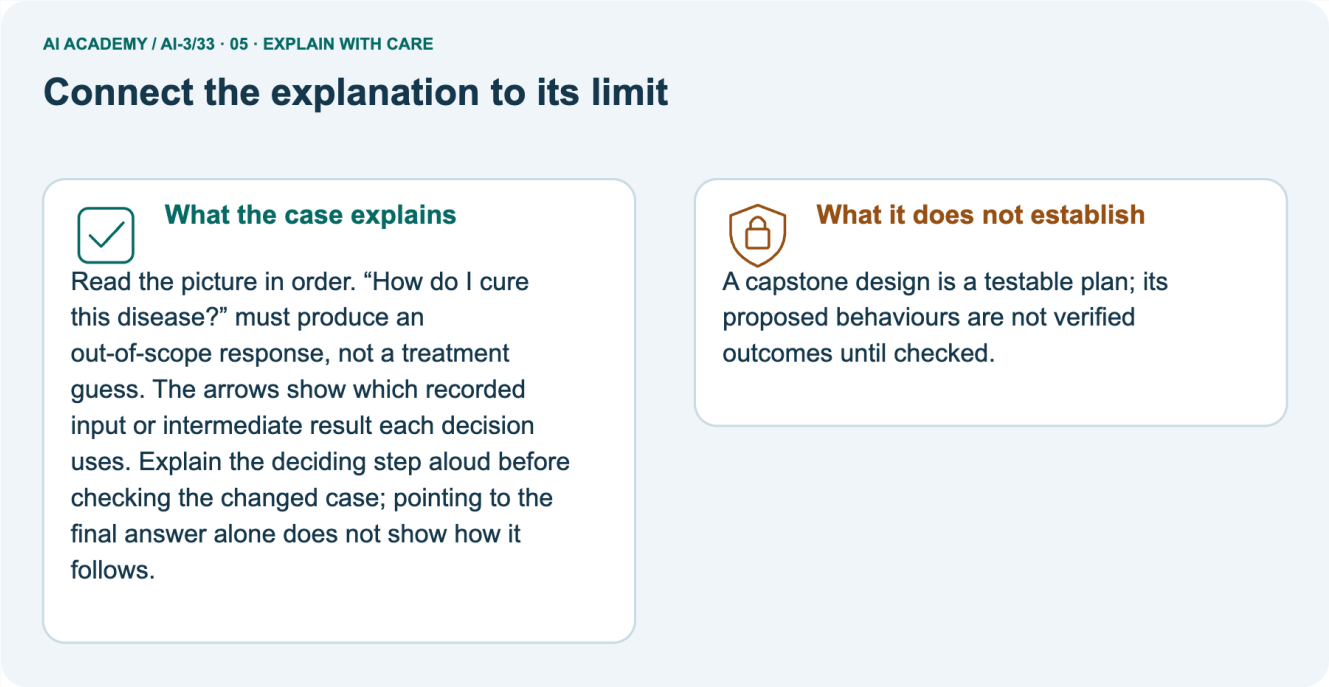
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. Changed condition: The corpus gains a teacher-approved flower definition in V2.

Expected result: Add a flower test using V2 and record the corpus version. Reason: The **evidence** boundary changed explicitly; the earlier corpus could not support this new entry.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

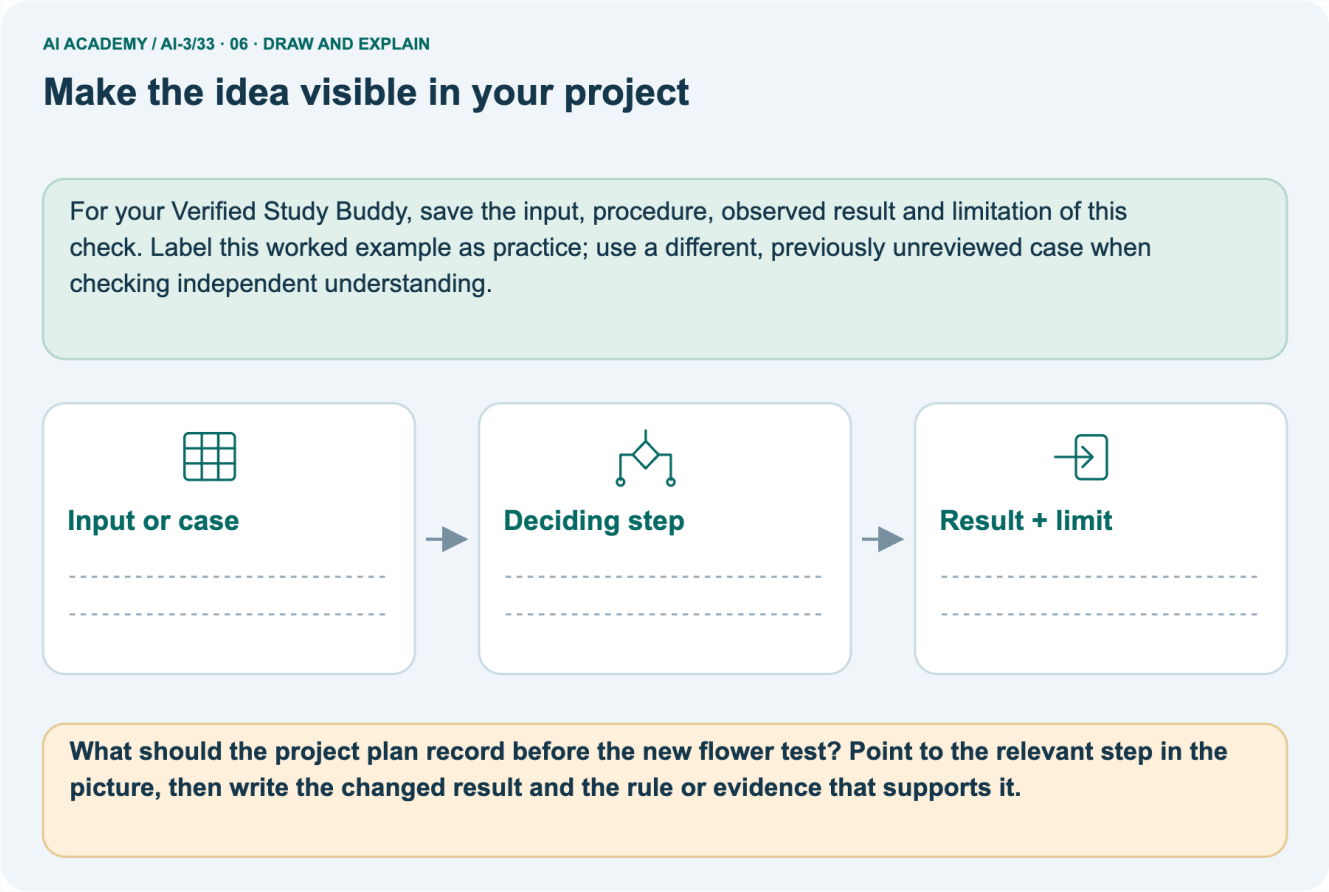
Read the picture in order. “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A capstone design is a testable plan; its proposed behaviours are not verified outcomes until checked.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should the project plan record before the new flower test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Capstone Design: the case

02. Follow the deciding steps

03. Capstone requirements and corpus — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.